

Benchmark-Relative Systematic Portfolio Construction: An Independent Ten-ETF Historical Study

Aniket Bhardwaj

9 October 2026

Independent research; not peer reviewed. Retrospective pseudo-out-of-sample evidence.

Abstract

This study examines how benchmark-relative tracking-error and turnover limits compress a fixed composite of momentum, low-volatility and short-term reversal preferences in a surviving ten-ETF panel. Monthly portfolios are compared with a monthly 60/40 policy, equal weight and inverse volatility, with a conservative two-session information lag. Development, validation and final holdout periods are separated, and the protocol and dataset are frozen before constructed holdout portfolio performance is examined. Negative outcomes and interrupted paths remain part of the evidence. After inspecting development/validation execution failures, formation reserves were introduced and the corrected specification was frozen before holdout. The 8% and 12% TE configurations have essentially identical performance; turnover was more restrictive under the tested settings, without establishing a general ranking of portfolio constraints. Exact-input reproduction requires the original, non-public market-data bundle; fresh retrieval supports approximate replication on a potentially revised vintage. Signal capture measures preference expression, not predictive skill. The results below distinguish arithmetic active returns, compounded performance and descriptive block-bootstrap uncertainty; they do not establish persistent alpha or causal effects of volatility regimes. In the final holdout, the primary model's net CAGR is 13.72% versus 12.33% for the frictionless policy. Annualized arithmetic active return is 1.23% with a descriptive 95% block interval [-1.55%, 3.90%]. Weekly execution failures in development and validation remain part of the robustness evidence.

1 Introduction and research questions

How do benchmark risk and trading constraints affect signal expression? Does increased expression improve net historical performance? Can a tighter tracking-error budget during high policy volatility materially change outcomes? The policy is a proposed comparison, not a claim of historical optimality. The original deterministic synthetic study remains unchanged and separate.

2 Literature review

Ledoit and Wolf motivate shrinkage risk estimation [4]; their covariance result does not establish profitability of this ETF strategy. DeMiguel, Garlappi and Uppal motivate a transparent equal-weight comparison [5]. Their conclusions concern different universes and estimation problems. Block resampling recognizes serial dependence rather than treating daily observations as independent [6]. This study applies those methodological ideas to portfolio construction without claiming a new asset-pricing discovery. Momentum research documents winner-loser persistence in individual stocks [1]; volatility-return research studies stock-level aggregate and idiosyncratic risk [2]; reversal research considers short-horizon contrarian stock returns [3]. These findings motivate preferences, not calibrated ETF expected returns. Total ETF volatility is not idiosyncratic stock volatility, and a 21-session reversal score is not a replication of a weekly stock-reversal study. Economic portability is an empirical question.

3 Historical dataset and data quality

The ordered universe is SPY, IWM, EFA, EEM, IEF, TLT, LQD, HYG, GLD and VNQ. Equity covers the first four; fixed income covers IEF, TLT, LQD and HYG; gold and real estate are separate constraint groups. VNQ is treated as a separate real-estate allocation despite its equity-like risk. Funds were selected retrospectively and survived to retrieval: this is not an unbiased historical investable universe. Every fund existed before January 2009; issuer inception and first-listing sources are recorded in the provenance appendix document. GLD’s November 18, 2004 availability date is its first trading date, distinct from the trust’s November 12 formation date.

The TigZig public API supplies Yahoo Finance data via yfinance. Its documented auto-adjusted Close incorporates splits and distributions [7]. The action ledger is preserved without a second adjustment. The operator’s terms describe educational/demonstration use and disclaim accuracy [8]; an institutional licence and upstream redistribution permission are not claimed. Raw responses remain outside Git. Adjusted returns approximate total returns with provider reinvestment conventions; tax and exact distribution reinvestment execution are not modelled. Latest-vintage revisions can differ from historical information available at each date.

Validation checks hashes, strict timestamp order, duplicates, positive finite prices, common NYSE sessions, per-fund availability, distributions/splits, 20% absolute daily-return review thresholds, original-response Close values, and actions from a second endpoint of the same provider. The second endpoint is not an independent market-data validation.

The panel contains 4,463 common sessions from January 2, 2009 to September 30, 2026, with 0 missing sessions and 0 threshold exceedances. The ledger contains 1,133 distributions and no splits in this window; GLD has no cash distributions.

4 Mathematical model

With score s , benchmark b , annual covariance Σ and known lagged holdings w^{known} , the existing convex program solves

$$\max_w s^\top w - 10(w - b)^\top \Sigma (w - b) - 0.1 \|w - w^{\text{known}}\|_1.$$

Fully invested long-only portfolios satisfy $\mathbf{1}^\top w = 1$ and $0 \leq w_i \leq 0.35$. At execution, annual estimated TE is at most 8% (5% in the high-state alternative), absolute active group exposure is at most 10 percentage points, and $T = \frac{1}{2} \|w - w^{\text{pretrade}}\|_1 \leq 0.30$. Every accepted solution is independently checked. Infeasible or inaccurate solver outcomes stop a path rather than silently relaxing constraints.

At return date t , signal and decision information end at $t - 2$; target weights are formed then, modelled at close $t - 1$, and earn returns after that close. Execution-close drift changes actual turnover but does not change target formation. Holdings drift as

$$w_{i,t}^{\text{end}} = \frac{w_{i,t}(1 + r_{i,t})}{1 + w_t^\top r_t}, \quad R_t^{\text{net}} = w_t^\top r_t - \frac{c}{10000} T_t.$$

The additive fee convention is a declared NAV approximation; this does not prove closing prices were executable. Exact share/cash accounting is omitted.

5 Signals and covariance methodology

For information date u , raw momentum is $P_{u-21}/P_{u-252} - 1$; low volatility is the negative sample standard deviation of the last 63 daily returns; reversal is $-(P_u/P_{u-21} - 1)$. Cross-sectional 5th/95th winsorization and population-standard-deviation standardization precede the composite $s = z_{\text{momentum}} + z_{\text{lowvol}} + 0.5z_{\text{reversal}}$. Coefficients are fixed preferences, not calibrated expected returns. Covariance uses 252

trailing returns, Ledoit–Wolf shrinkage, a 10^{-10} daily diagonal ridge and annualization 252. Sample covariance is a declared robustness alternative. Forward IC is Spearman correlation with subsequent asset returns through the next rebalance; the terminal period can be partial.

Signal capture is $(w - b)^\top s$ divided by the expression of a same-score, same-information long-only reference with the 35% position cap but without TE, group, turnover constraints or penalties. Exposure is measured against the realized execution benchmark for both paths. Absolute reference exposure at most 10^{-10} yields an undefined observation. Different ablations have different reference scores, so their capture ratios are not cardinal skill comparisons. High capture does not imply superior returns.

6 Policy benchmark and constraints

The policy weights, in universe order, are (0.30, 0.05, 0.15, 0.10, 0.20, 0.05, 0.15, 0, 0, 0). It resets at the close immediately before the first observed session return of each month and drifts between resets. The schedule is independently represented when the active strategy rebalances weekly. Ordinary policy, equal-weight and inverse-volatility comparisons satisfy their own rules and portfolio accounting; active-manager TE, sector and turnover mandates are not imposed on them. Both frictionless and independently cost-adjusted policy returns are exported. Each independent period starts endowed at its execution-date policy holdings; the active transition is charged and the initial policy endowment is not charged.

6.1 Pre-holdout design revision after execution failures

Original development/validation active paths without reserves failed execution checks, predominantly turnover limits following unobserved one-session drift; those failures remain archived. After inspecting these development/validation outcomes, Codex introduced identical formation reserves across active scenarios: 2 percentage points of turnover, 1 percentage point of group exposure and 0.1 percentage point of annual TE. Thus primary formation caps are 28%, 9 percentage points and 7.9%; the high-state formation TE cap is 4.9%. Execution mandates remain 30%, 10 percentage points and 8%/5%, respectively. This is an outcome-informed engineering correction to the initial specification. The corrected specification was reviewed and frozen before constructed holdout portfolio performance was examined; published results use that corrected model. Reserves address engineering feasibility and do not guarantee that execution constraints remain feasible. Remaining weekly sector failures in development and validation were retained. A benchmark target fits the 35% cap; transition constraints can still be infeasible.

7 Experiment design and holdout protocol

Warmup is January 2009–December 2010; development is January 3, 2011–December 29, 2017; validation is January 2, 2018–December 30, 2022; final holdout is January 3, 2023–September 30, 2026. Models receive physical price slices ending at each period boundary. Before holdout, data checks, tests, development/validation review and the configuration were frozen with dataset hashes and a code commit. An exclusive access ledger was written before the single final evaluation. No favourable ablation was substituted for the primary model. Inherited source discovery and issuer verification exposed incidental holdout-era quotes/performance panels, and full prices were ingested for quality checks; constructed holdout performance was not used for selection. The historical split is retrospective, not prospective preregistration.

Experiments A–G cover TE caps 4/8/12%; turnover caps 10/30/50%; identical-trade cost sensitivities 0/10/25/40 bps; fixed 8% versus normal 8%/high 5% TE; three signal omissions plus regime awareness; the five portfolio strategies; and sample covariance, three-session information delay and weekly rebalancing. The full ledgers retain failures. Cost assumptions do not reoptimize holdings. Metrics use 252

observations/year and sample return standard deviations; Sharpe/Sortino assume zero risk-free return. Undefined metrics are explicit.

Uncertainty uses circular 21-session blocks, 2,000 resamples, seed 20261008 and 95% percentile intervals for annualized arithmetic active means. Strategy differences are paired on the same dates. Approximate within-period stationarity and dependence within blocks are assumed; structural breaks, longer regime persistence and multiple comparisons are not resolved. No naive independent-day significance tests or persistent-alpha conclusions are made.

8 Empirical performance results

All returns below are computed from executed paths. CAGR, annualized arithmetic active return and TE are distinct quantities. Active returns use the frictionless policy; a cost-adjusted policy comparison is included in the machine-readable tables. Percent figures are expressed in percentage points.

8.1 Development

Strategy	Gross CAGR	Net CAGR	TE	Active	MDD
Policy (net)	7.987	7.976	0.004	-0.011	-11.373
Equal weight	6.879	6.859	2.462	-1.082	-11.692
Inverse volatility	6.439	6.384	4.355	-1.697	-9.245
Fixed risk	7.722	7.499	3.653	-0.380	-17.784
Regime aware	7.710	7.486	3.626	-0.396	-17.440

Fixed-risk arithmetic active return is -0.38% with a 95% block interval [-3.14%, 2.08%]. Mean capture is 0.665, with 84 valid and 0 excluded rebalance observations. Mean forward IC is 0.012. Frictionless policy CAGR is 7.99%; independently cost-adjusted policy CAGR is 7.98%.

8.2 Validation

Strategy	Gross CAGR	Net CAGR	TE	Active	MDD
Policy (net)	3.284	3.271	0.004	-0.013	-24.271
Equal weight	2.766	2.743	2.752	-0.618	-24.064
Inverse volatility	1.461	1.400	5.436	-2.228	-22.547
Fixed risk	2.283	2.033	3.294	-1.148	-25.315
Regime aware	2.330	2.086	3.250	-1.093	-25.315

Fixed-risk arithmetic active return is -1.15% with a 95% block interval [-3.67%, 1.36%]. Mean capture is 0.548, with 60 valid and 0 excluded rebalance observations. Mean forward IC is -0.060. Frictionless policy CAGR is 3.28%; independently cost-adjusted policy CAGR is 3.27%.

8.3 Holdout

Strategy	Gross CAGR	Net CAGR	TE	Active	MDD
Policy (net)	12.325	12.315	0.003	-0.009	-9.729
Equal weight	11.405	11.379	2.841	-0.853	-9.785
Inverse volatility	9.093	9.034	3.746	-3.160	-8.347
Fixed risk	13.980	13.717	2.782	1.233	-11.089
Regime aware	13.964	13.700	2.773	1.218	-11.089

Fixed-risk arithmetic active return is 1.23% with a 95% block interval [-1.55%, 3.90%]. Mean capture is 0.730, with 45 valid and 0 excluded rebalance observations. Mean forward IC is -0.019. Frictionless policy CAGR is 12.33%; independently cost-adjusted policy CAGR is 12.32%.

9 Constraint sensitivity findings

Table 2 reports every declared scenario. Relaxing a risk cap need not change an allocation when another constraint or the objective dominates. The 8% and 12% TE configurations have essentially identical performance, consistent with largely non-binding TE caps at those settings. Turnover restricted signal expression more than the tested 4%–12% TE range under this model, objective, universe and other constraints. This conditional sensitivity does not establish that turnover is generally more important than tracking error in portfolio management. Higher capture does not establish better returns. Capture and turnover frontiers describe construction mechanics rather than investment skill. Formation reserves apply identically across frontier scenarios; 10/30/50% turnover mandates imply 8/28/48% formation caps.

10 Volatility-regime comparison

The classifier uses the monthly-policy return history: trailing 63-session sample volatility times $\sqrt{252}$, compared with the expanding 75th percentile of strictly earlier valid rolling observations, requiring 252 earlier values. Only strict exceedances are high; ties are normal. Labels used in allocation are dated at the lagged information close and budgets change at rebalances. The timeline shows observation-date labels, whose availability is after that close. Conditional results use daily lagged labels and arithmetic summaries; they do not compound discontinuous subsequences or imply regime causality.

Development: regime-aware minus fixed annualized arithmetic return -0.02%; paired 95% block interval [-0.12%, 0.10%].

Validation: regime-aware minus fixed annualized arithmetic return 0.05%; paired 95% block interval [-0.07%, 0.18%].

Holdout: regime-aware minus fixed annualized arithmetic return -0.02%; paired 95% block interval [-0.10%, 0.05%].

Period	Strategy	State	Days	Active	TE
development	Fixed risk	normal	1607	-0.561	3.241
development	Fixed risk	high	154	1.507	6.577
development	Regime aware	normal	1607	-0.581	3.246
development	Regime aware	high	154	1.532	6.372
validation	Fixed risk	normal	810	-0.755	2.786
validation	Fixed risk	high	449	-1.855	4.056
validation	Regime aware	normal	810	-0.686	2.767
validation	Regime aware	high	449	-1.827	3.979
holdout	Fixed risk	normal	722	1.506	2.545
holdout	Fixed risk	high	217	0.328	3.461
holdout	Regime aware	normal	722	1.512	2.541
holdout	Regime aware	high	217	0.241	3.439

11 Robustness and ablation results

The alternative covariance, delayed-information and signal-omission experiments remain separately labelled. They are descriptive checks, not an uncontrolled parameter search. Weekly paths that breach execution constraints receive no full-period CAGR or fabricated continuation. The omission plots preserve the full original model rather than selecting the best historical omission.

12 Economic interpretation

Costs mechanically reduce net wealth on an unchanged trade path. A stronger preference alignment can coexist with underperformance, because standardized signal scores are not expected-return forecasts. Changes in relative performance across historical periods can reflect asset-class composition, macroeconomic exposures, estimator error and noise. No regression-based risk-adjusted alpha, causal regime effect or investable closing-auction advantage is established.

13 Limitations

The surviving universe, latest adjustment vintage and Yahoo-derived data are material selection and measurement limitations. ETF prices embed fund expenses; taxes, share-lot/cash execution, market impact, liquidity, funding and settlement are omitted. Turnover is half-L1 and fees are additive fractions of beginning NAV. Constraints hold at accepted rebalances; drift can violate limits between them, and realized TE can exceed ex-ante caps. Fixed risk reserves cannot guarantee future mandate compliance. The modest fixed panel and multiple comparisons limit generalization; block intervals do not solve structural-break uncertainty. Zero-rate Sharpe is not a comparison with historical cash yields. The split is retrospective and independently reproduced source prices can change vintage. The original market-data bundle is retained locally and is not publicly available. Fingerprints establish identity but cannot supply those inputs; exact external reproduction is therefore conditional on original-bundle access. Fresh downloads can support method replication while changing numerical findings.

14 Conclusion

The empirical workflow provides auditable evidence of portfolio-construction tradeoffs under declared benchmark risk, trading and cost conventions. It also preserves failed transitions and inconclusive uncertainty. Under the tested model and parameter range, turnover restricted signal expression more than the TE range, whose 8% and 12% caps were largely non-binding. This is not a general ranking of portfolio constraints. The reserves were an engineering correction informed by development/validation failures and frozen before holdout. Signal expression is a construction diagnostic; these historical results do not establish persistent alpha. Prospective evaluation and independently licensed institutional data would strengthen further research.

Research figures

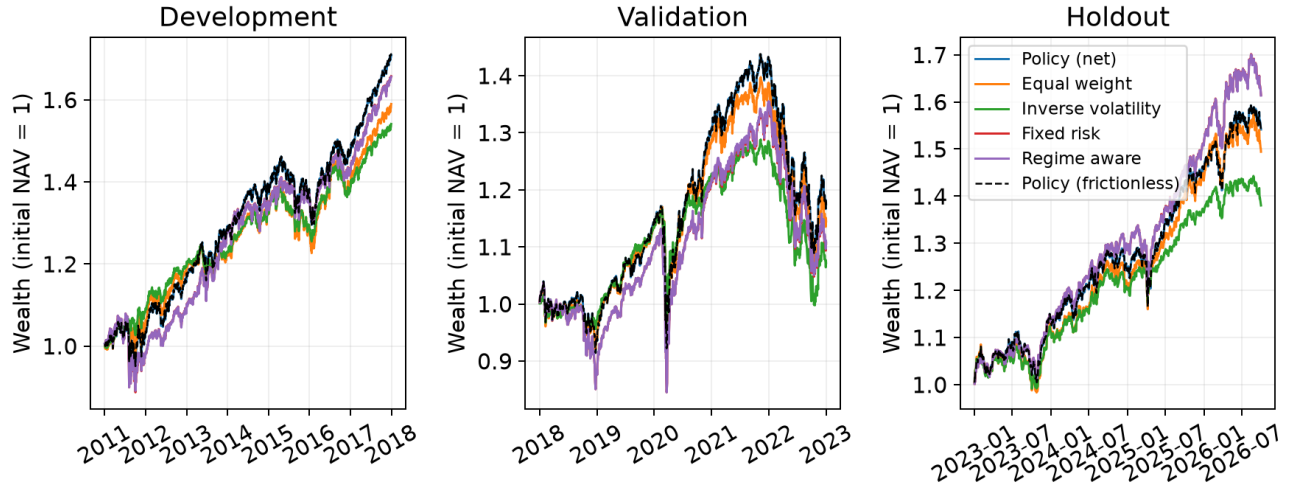


Figure 1: Executed cumulative net wealth with the frictionless policy shown separately. Each period is independently endowed at NAV 1.

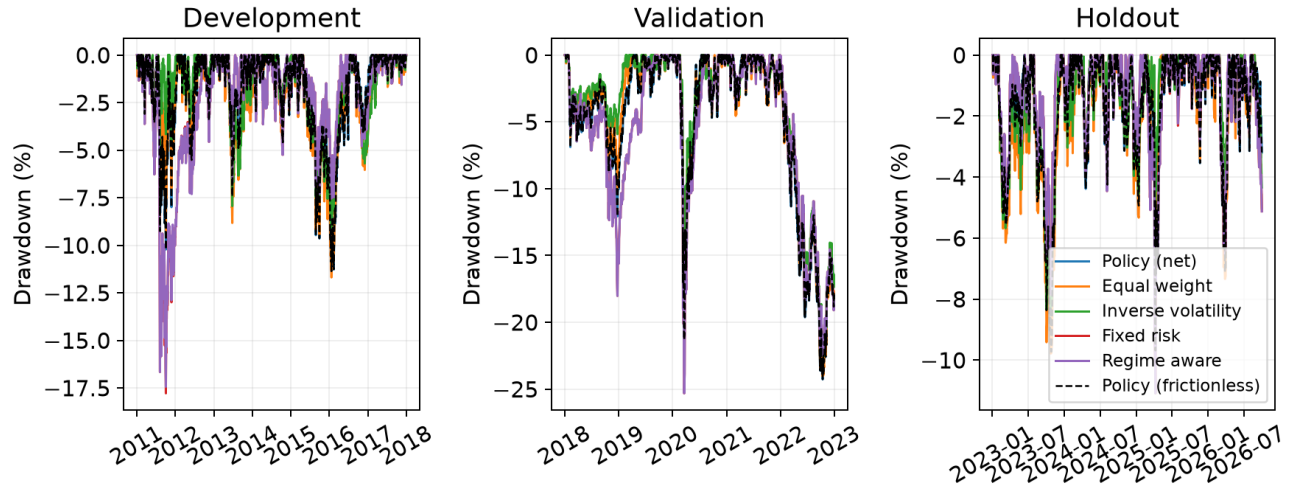


Figure 2: Executed net drawdowns, including initial NAV 1 in the running high-water mark.

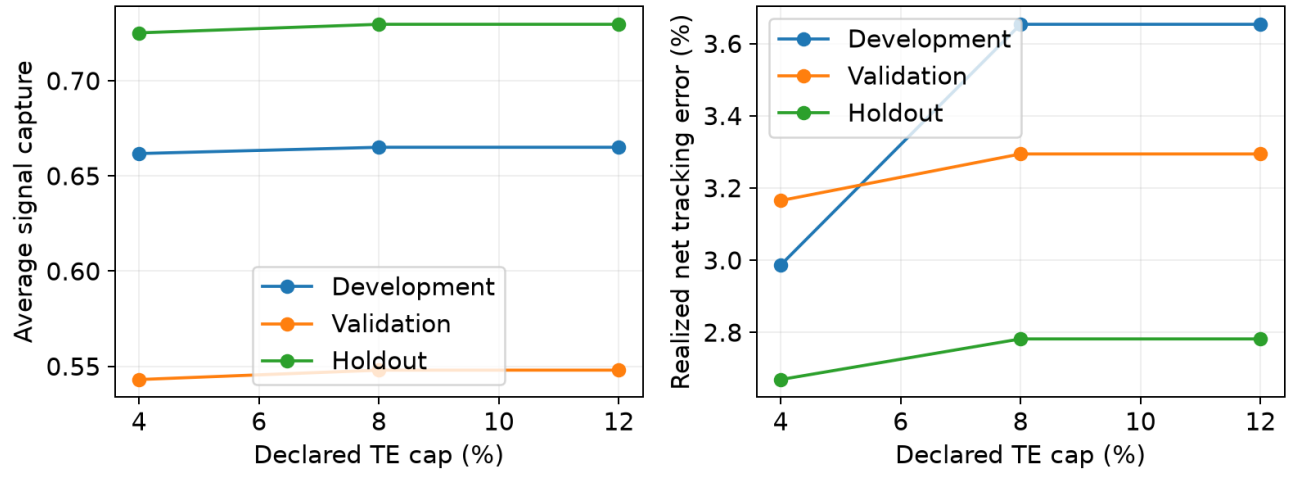


Figure 3: Declared TE mandates versus average signal capture and realized net TE; all other primary settings fixed.

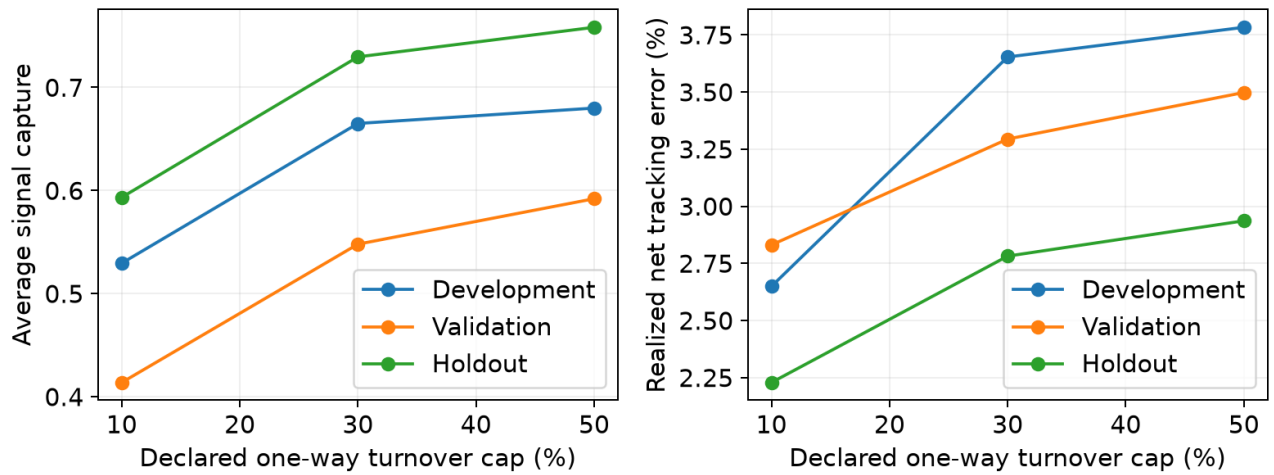


Figure 4: Declared turnover mandates versus capture and realized net TE; the identical two-percentage-point formation reserve applies throughout.

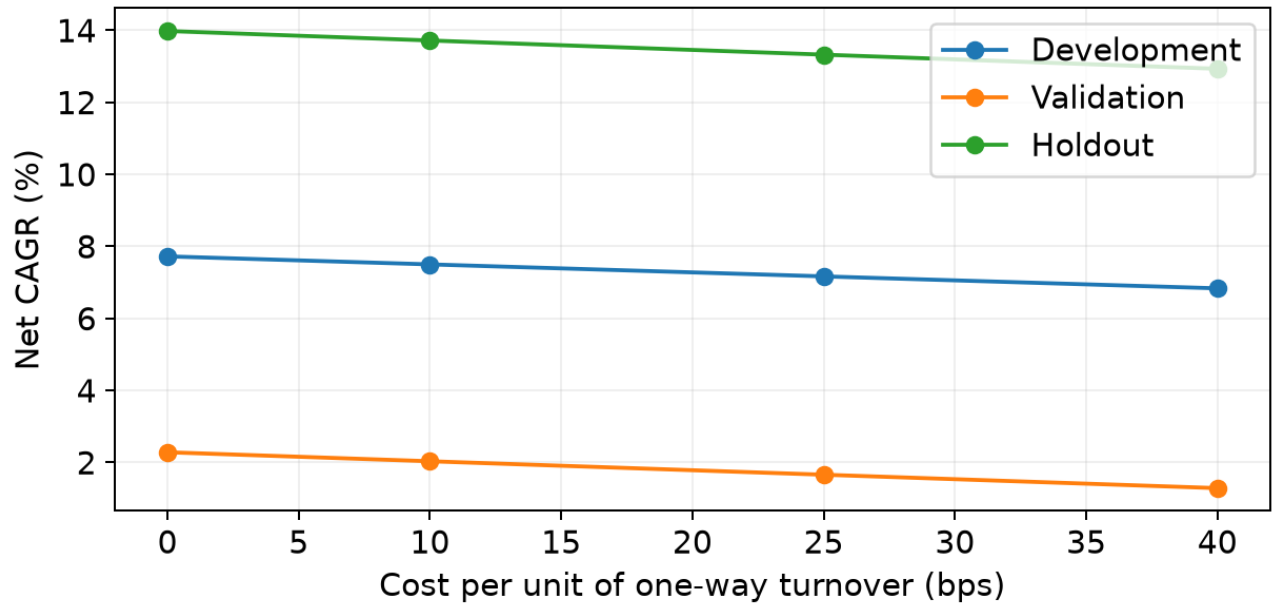


Figure 5: Net CAGR under accounting-only cost changes with identical gross returns, holdings and trades.

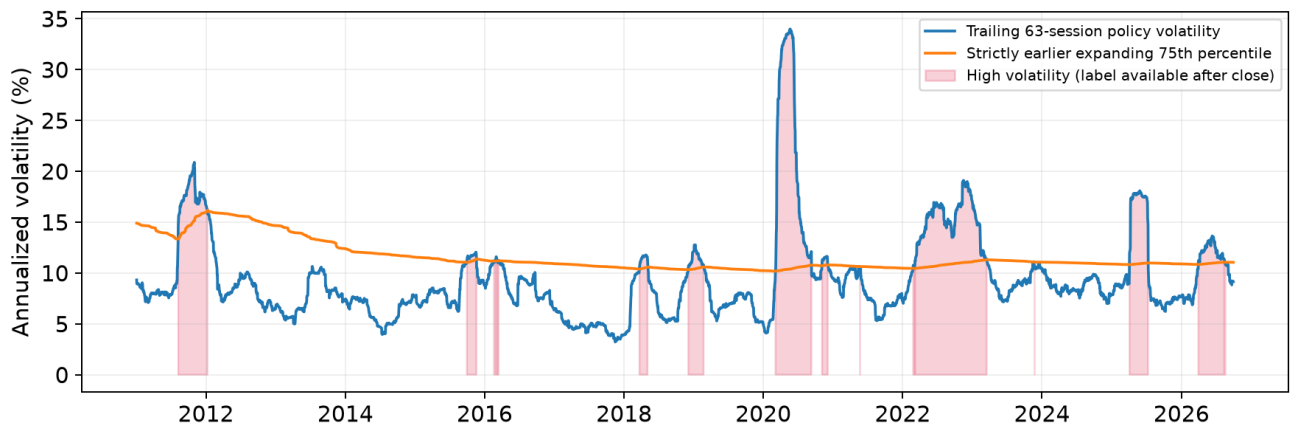


Figure 6: Policy volatility and its strictly earlier expanding threshold. Red shading marks observation-date high-state labels, available only after that close.

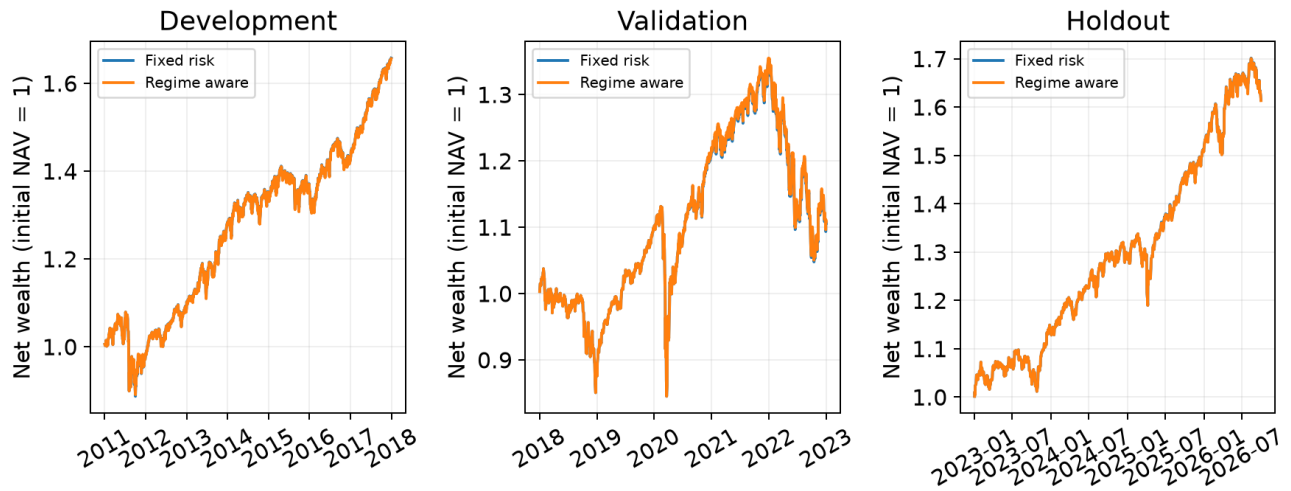


Figure 7: Executed fixed-risk and regime-aware net wealth; small visual differences need not indicate meaningful skill.

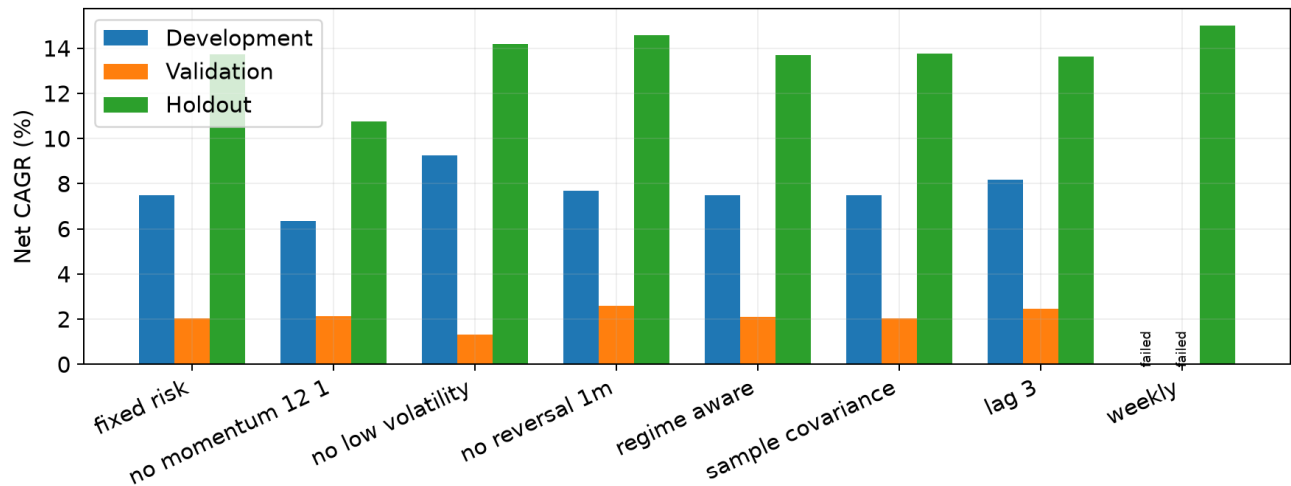


Figure 8: Executed ablations and robustness comparisons. Failed full-period paths have no return bar.

References

- [1] Jegadeesh, N., and Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance* 48(1), 65–91.
<https://doi.org/10.1111/j.1540-6261.1993.tb04702.x>.
- [2] Ang, A., Hodrick, R. J., Xing, Y., and Zhang, X. (2006). The cross-section of volatility and expected returns. *Journal of Finance* 61(1), 259–299.
<https://doi.org/10.1111/j.1540-6261.2006.00836.x>.
- [3] Lehmann, B. N. (1990). Fads, martingales, and market efficiency. *Quarterly Journal of Economics* 105(1), 1–28. <https://doi.org/10.2307/2937816>.
- [4] Ledoit, O., and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis* 88(2), 365–411.
[https://doi.org/10.1016/S0047-259X\(03\)00096-4](https://doi.org/10.1016/S0047-259X(03)00096-4).
- [5] DeMiguel, V., Garlappi, L., and Uppal, R. (2009). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Review of Financial Studies* 22(5), 1915–1953.
<https://doi.org/10.1093/rfs/hhm075>.
- [6] Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics* 17(3), 1217–1241. <https://doi.org/10.1214/aos/1176347265>.
- [7] TigZig (2026). Yahoo Finance Data API documentation.
<https://www.tigzig.com/apis/yahoo-finance>. Accessed October 9, 2026.
- [8] TigZig (2026). Terms of Use (August 2026 revision). <https://www.tigzig.com/terms>. Accessed October 9, 2026.

A All declared scenario results

Table values are percentages for CAGR, active return, TE and turnover; capture is unitless. A dash denotes an unavailable full-period metric. Complete Sharpe, Sortino, volatility, drawdown, information ratio, wealth, cost drag, active share, IC and constraint diagnostics are in `all_performance.csv` and the per-scenario ledgers.

Table 2: All executed scenarios; failures are retained.

Period	Scenario	Net CAGR	Active	TE	Capture	Turnover
development	fixed_risk	7.499	-0.380	3.653	0.665	17.210
development	te_4pct	7.790	-0.168	2.986	0.662	17.273
development	te_8pct	7.499	-0.380	3.653	0.665	17.210
development	te_12pct	7.499	-0.380	3.653	0.665	17.210
development	turnover_10pct	8.003	0.025	2.649	0.529	7.272
development	turnover_30pct	7.499	-0.380	3.653	0.665	17.210
development	turnover_50pct	7.532	-0.345	3.783	0.680	20.522
development	cost_0bps	7.722	-0.173	3.651	0.665	17.210
development	cost_10bps	7.499	-0.380	3.653	0.665	17.210
development	cost_25bps	7.166	-0.690	3.658	0.665	17.210
development	cost_40bps	6.834	-1.001	3.666	0.665	17.210
development	regime_aware	7.486	-0.396	3.626	0.665	17.316
development	without_momentum_12_1	6.351	-1.581	2.386	0.419	17.198
development	without_low_volatility	9.256	1.402	4.989	0.629	21.105
development	without_reversal_1m	7.678	-0.293	2.837	0.689	10.438
development	policy	7.976	-0.011	0.004	–	0.885
development	equal_weight	6.859	-1.082	2.462	–	1.557

Period	Scenario	Net CAGR	Active	TE	Capture	Turnover
development	inverse_volatility	6.384	-1.697	4.355	–	4.241
development	sample_covariance	7.494	-0.384	3.653	0.665	17.245
development	lag_3	8.178	0.163	2.963	0.653	17.595
development	weekly	–	–	–	–	–
validation	fixed_risk	2.033	-1.148	3.294	0.548	20.398
validation	te_4pct	2.203	-0.979	3.165	0.543	19.746
validation	te_8pct	2.033	-1.148	3.294	0.548	20.398
validation	te_12pct	2.033	-1.148	3.294	0.548	20.398
validation	turnover_10pct	2.451	-0.718	2.831	0.414	7.816
validation	turnover_30pct	2.033	-1.148	3.294	0.548	20.398
validation	turnover_50pct	2.692	-0.502	3.498	0.592	27.121
validation	cost_0bps	2.283	-0.903	3.291	0.548	20.398
validation	cost_10bps	2.033	-1.148	3.294	0.548	20.398
validation	cost_25bps	1.659	-1.515	3.301	0.548	20.398
validation	cost_40bps	1.286	-1.883	3.313	0.548	20.398
validation	regime_aware	2.086	-1.093	3.250	0.546	19.938
validation	without_momentum_12_1	2.143	-1.108	3.547	0.385	19.809
validation	without_low_volatility	1.312	-1.765	4.989	0.576	20.635
validation	without_reversal_1m	2.591	-0.642	3.208	0.604	12.233
validation	policy	3.271	-0.013	0.004	–	1.112
validation	equal_weight	2.743	-0.618	2.752	–	1.881
validation	inverse_volatility	1.400	-2.228	5.436	–	5.065
validation	sample_covariance	2.032	-1.148	3.293	0.548	20.375
validation	lag_3	2.463	-0.745	3.276	0.534	21.237
validation	weekly	–	–	–	–	–
holdout	fixed_risk	13.717	1.233	2.782	0.730	19.135
holdout	te_4pct	13.231	0.786	2.669	0.725	18.822
holdout	te_8pct	13.717	1.233	2.782	0.730	19.135
holdout	te_12pct	13.717	1.233	2.782	0.730	19.135
holdout	turnover_10pct	14.216	1.639	2.228	0.593	7.338
holdout	turnover_30pct	13.717	1.233	2.782	0.730	19.135
holdout	turnover_50pct	13.508	1.040	2.936	0.758	22.762
holdout	cost_0bps	13.980	1.465	2.782	0.730	19.135
holdout	cost_10bps	13.717	1.233	2.782	0.730	19.135
holdout	cost_25bps	13.323	0.887	2.785	0.730	19.135
holdout	cost_40bps	12.931	0.540	2.792	0.730	19.135
holdout	regime_aware	13.700	1.218	2.773	0.729	19.198
holdout	without_momentum_12_1	10.752	-1.481	2.518	0.423	16.535
holdout	without_low_volatility	14.202	1.891	4.250	0.594	20.959
holdout	without_reversal_1m	14.577	2.005	2.951	0.807	10.956
holdout	policy	12.315	-0.009	0.003	–	0.762
holdout	equal_weight	11.379	-0.853	2.841	–	1.886
holdout	inverse_volatility	9.034	-3.160	3.746	–	4.468
holdout	sample_covariance	13.762	1.273	2.785	0.730	19.142
holdout	lag_3	13.640	1.140	2.777	0.738	19.189
holdout	weekly	14.992	2.377	3.207	0.747	10.037

Failure: development, weekly: Execution mandate failed on 2012-06-04: sector active weight, sector active weight.

Failure: validation, weekly: Execution mandate failed on 2020-03-16: sector active weight.

B Reproduction appendix

Frozen code commit: 86a72b72584cf70e990525b62a7d4fab6e3649bd.

Dataset SHA-256: 5af0b0c65a9b4f8be569090f974c0b5446839a0bac20c3485d2184109cf981ea.

See docs/empirical_phase2.md for the data-provider and issuer sources, environment versions, original failures, exact commands and verification record.

Exact-input reproduction. The original bundle (original responses, prices, actions and manifest) is local and not publicly distributed. Exact-input reproduction requires access to that bundle, matching

recorded fingerprints, the frozen model commit and protocol, and compatible recorded dependencies. Numerical agreement is subject to solver tolerances; byte-for-byte equality across environments is not promised. The local validation reproduction and accounting audit used the original bundle; they do not establish that an external reader can retrieve the same vintage. Dataset hashes identify the inputs but cannot provide them.

Approximate replication with a new vintage. The acquisition commands retrieve fresh provider data, whose historical adjusted prices or actions may have changed. This replicates the method, not necessarily the published numerical results. Preserve a new manifest and fingerprints and use a new explicitly labelled output directory. The original frozen holdout output must not be overwritten or retuned. Public ledgers, tables and figures support evidence inspection without supplying independent source-price reconstruction. The reporting script reads executed ledgers and never calls the optimizer. The manuscript compiles from the repository root using the documented Tectonic command.